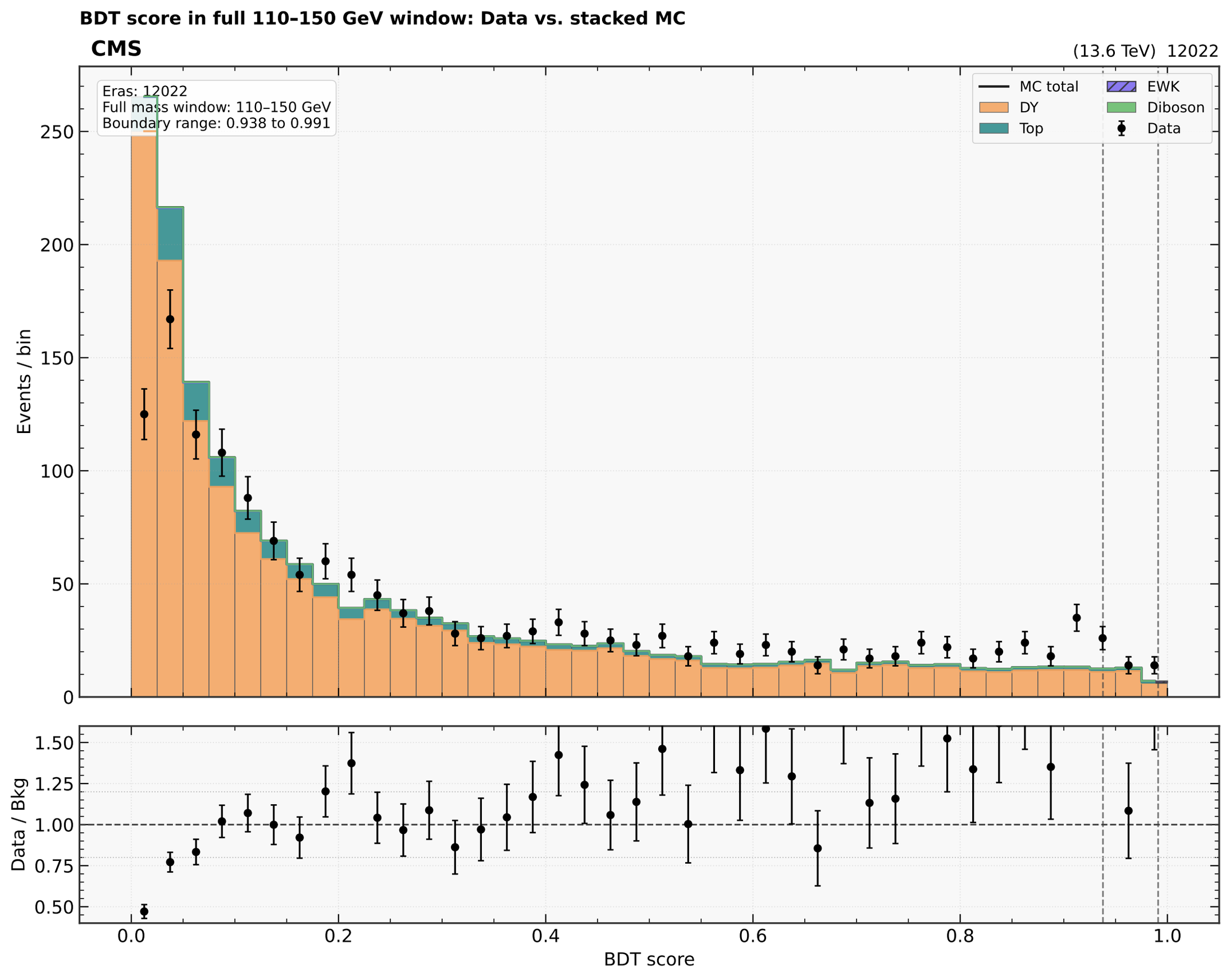
Run-3 H → μμ VBF BDT Category Optimisation

Eras : 12022
Category edges : 0.0000, 0.9378, 0.9850, 0.9910, 1.0000
MC scale factor : 1.0000 (Data/MC sideband normalisation)
Blinding window : 115–135 GeV
Sidebands : (110,115) ∪ (135,150) GeV
Outputs dir : /eos/home-m/memmad/HiggstoDimuon_VBF/src/drivers/9_TMVA/Outputs
Extra selection : –

CONTENTS:

p.2 BDT score distribution: Data vs. MC
p.3 BDT bin yields
p.4 Significance vs. number of categories
p.5 Boundary scan step-by-step (AN Fig. 32)
p.6 Atanh BDT + S/√B per category
p.7 BDT output vs. mass hypothesis (AN Fig. 26R)
p.8 Signal FWHM vs. BDT score
p.9 ROC curve with AUC
p.10 Blinded dimuon mass spectrum
p.11 Mass sculpting check (AN Fig. 25)
p.12 Per-category mass distributions and background fits
p.13 Background model comparison (χ²/ndf)
p.14 VBF signal DCB fits per category
p.15 DCB parameter table
p.16 σ_CB vs. category
p.17 Signal acceptance × efficiency
p.18 Per-category yield and sensitivity summary
p.19 Signal mass shapes mH=120/125/130 (AN Fig. 40)
p.20 Category summary table
p.21 Category process composition

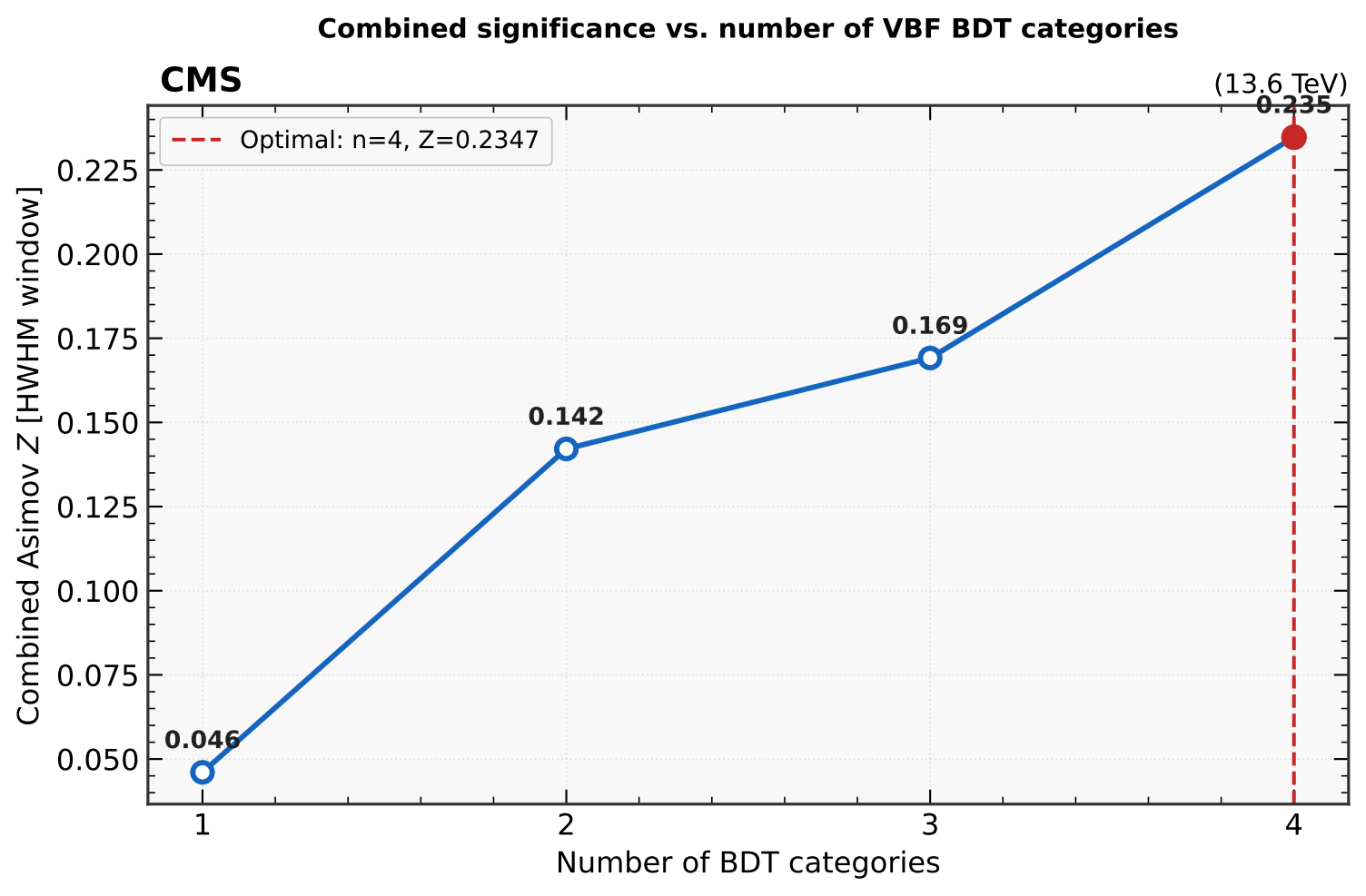

BDT score in full 110–150 GeV window: Data vs. stacked MC
CMS
(13.6 TeV) 12022
Eras: 12022
Full mass window: 110–150 GeV
Boundary range: 0.938 to 0.991
MC total
DY
Top
EWK
Diboson
Data
Events / bin
Data / Bkg
BDT score
0
50
100
150
200
250
0.50
0.75
1.00
1.25
1.50
0.0
0.2
0.4
0.6
0.8
1.0

BDT bin summary: signal, background, and sensitivity (page 1 / 2)

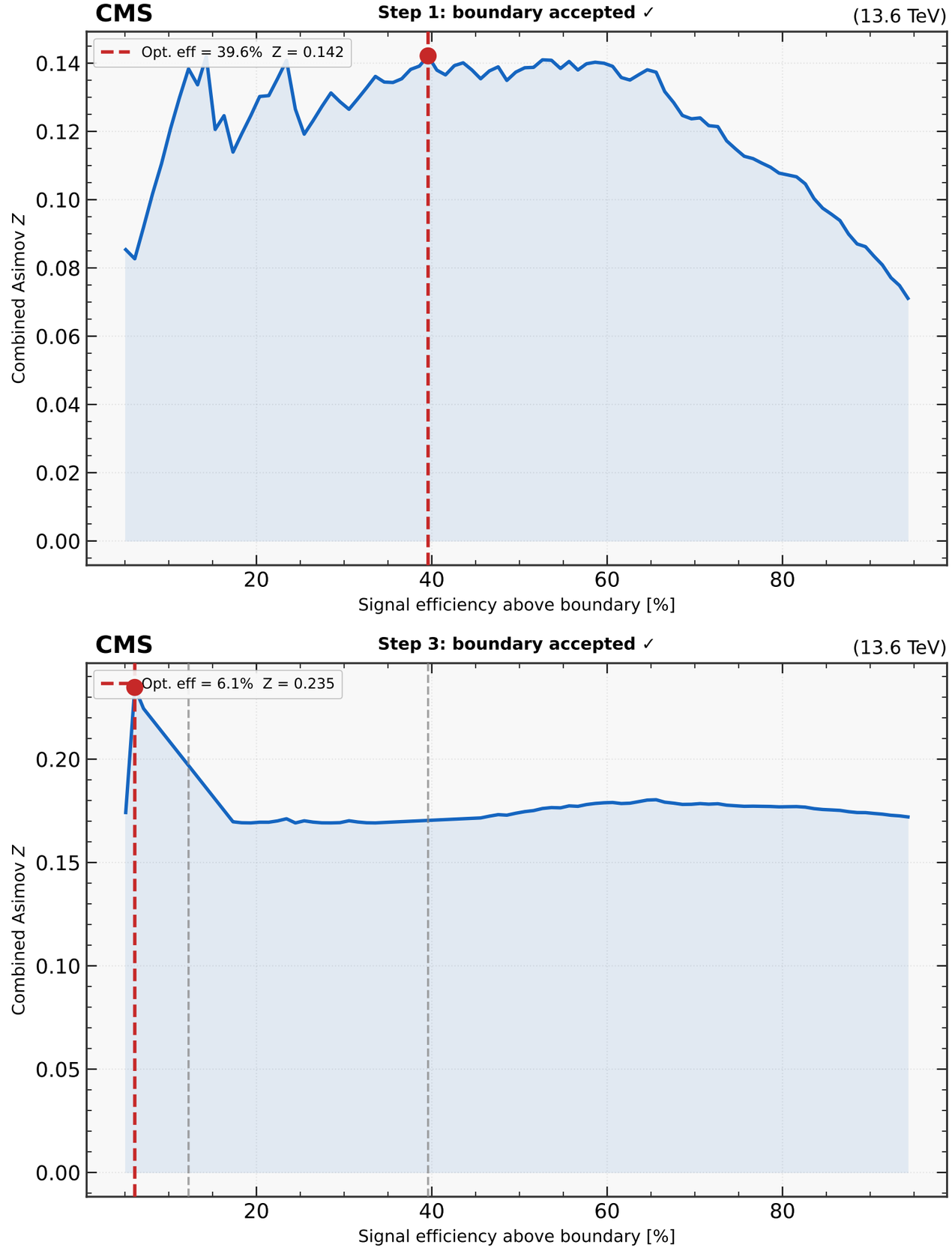
BDT bin	Total signal	Bkg. ± ΔB	Data	S/(S+B) (%)
[0.000, 0.025)	0.007	265.49 ± 6.86	125	0.00
[0.025, 0.050)	0.016	216.45 ± 6.17	167	0.01
[0.050, 0.075)	0.016	139.18 ± 4.94	116	0.01
[0.075, 0.100)	0.016	105.89 ± 4.36	108	0.02
[0.100, 0.125)	0.015	82.21 ± 3.84	88	0.02
[0.125, 0.150)	0.015	69.05 ± 3.53	69	0.02
[0.150, 0.175)	0.016	58.63 ± 3.27	54	0.03
[0.175, 0.200)	0.016	49.90 ± 3.02	60	0.03
[0.200, 0.225)	0.017	39.31 ± 2.64	54	0.04
[0.225, 0.250)	0.016	43.19 ± 2.81	45	0.04
[0.250, 0.275)	0.017	38.27 ± 2.70	37	0.04
[0.275, 0.300)	0.017	34.94 ± 2.56	38	0.05
[0.300, 0.325)	0.017	32.47 ± 2.48	28	0.05
[0.325, 0.350)	0.018	26.79 ± 2.15	26	0.07
[0.350, 0.375)	0.017	25.85 ± 2.20	27	0.07
[0.375, 0.400)	0.018	24.82 ± 2.16	29	0.07
[0.400, 0.425)	0.020	23.17 ± 2.12	33	0.09
[0.425, 0.450)	0.020	22.54 ± 2.08	28	0.09
[0.450, 0.475)	0.021	23.62 ± 2.12	25	0.09
[0.475, 0.500)	0.022	20.21 ± 1.92	23	0.11

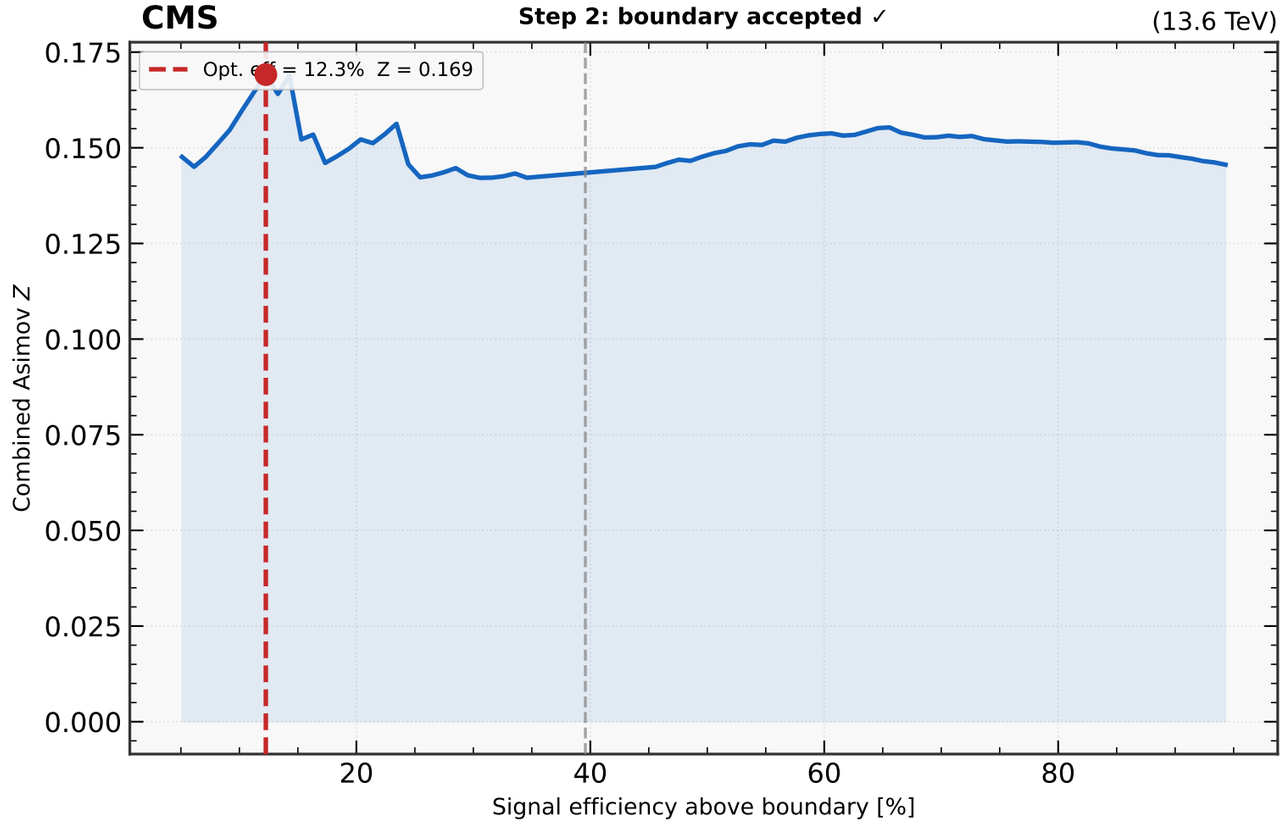
BDT bin summary: signal, background, and sensitivity (page 2 / 2)

BDT bin	Total signal	Bkg. ± ΔB	Data	S/(S+B) (%)
[0.500, 0.525)	0.023	18.48 ± 1.83	27	0.13
[0.525, 0.550)	0.026	17.94 ± 1.78	18	0.14
[0.550, 0.575)	0.025	14.51 ± 1.66	24	0.17
[0.575, 0.600)	0.028	14.27 ± 1.56	19	0.19
[0.600, 0.625)	0.031	14.52 ± 1.62	23	0.21
[0.625, 0.650)	0.033	15.46 ± 1.65	20	0.21
[0.650, 0.675)	0.036	16.36 ± 1.74	14	0.22
[0.675, 0.700)	0.038	11.97 ± 1.48	21	0.31
[0.700, 0.725)	0.044	15.02 ± 1.70	17	0.30
[0.725, 0.750)	0.047	15.55 ± 1.70	18	0.30
[0.750, 0.775)	0.054	14.08 ± 1.61	24	0.38
[0.775, 0.800)	0.060	14.43 ± 1.68	22	0.41
[0.800, 0.825)	0.068	12.71 ± 1.51	17	0.53
[0.825, 0.850)	0.078	12.36 ± 1.48	20	0.63
[0.850, 0.875)	0.096	13.09 ± 1.56	24	0.73
[0.875, 0.900)	0.118	13.32 ± 1.53	18	0.88
[0.900, 0.925)	0.148	13.28 ± 1.63	35	1.10
[0.925, 0.950)	0.204	12.49 ± 1.51	26	1.60
[0.950, 0.975)	0.325	12.91 ± 1.53	14	2.46
[0.975, 1.000)	0.476	7.05 ± 1.14	14	6.33

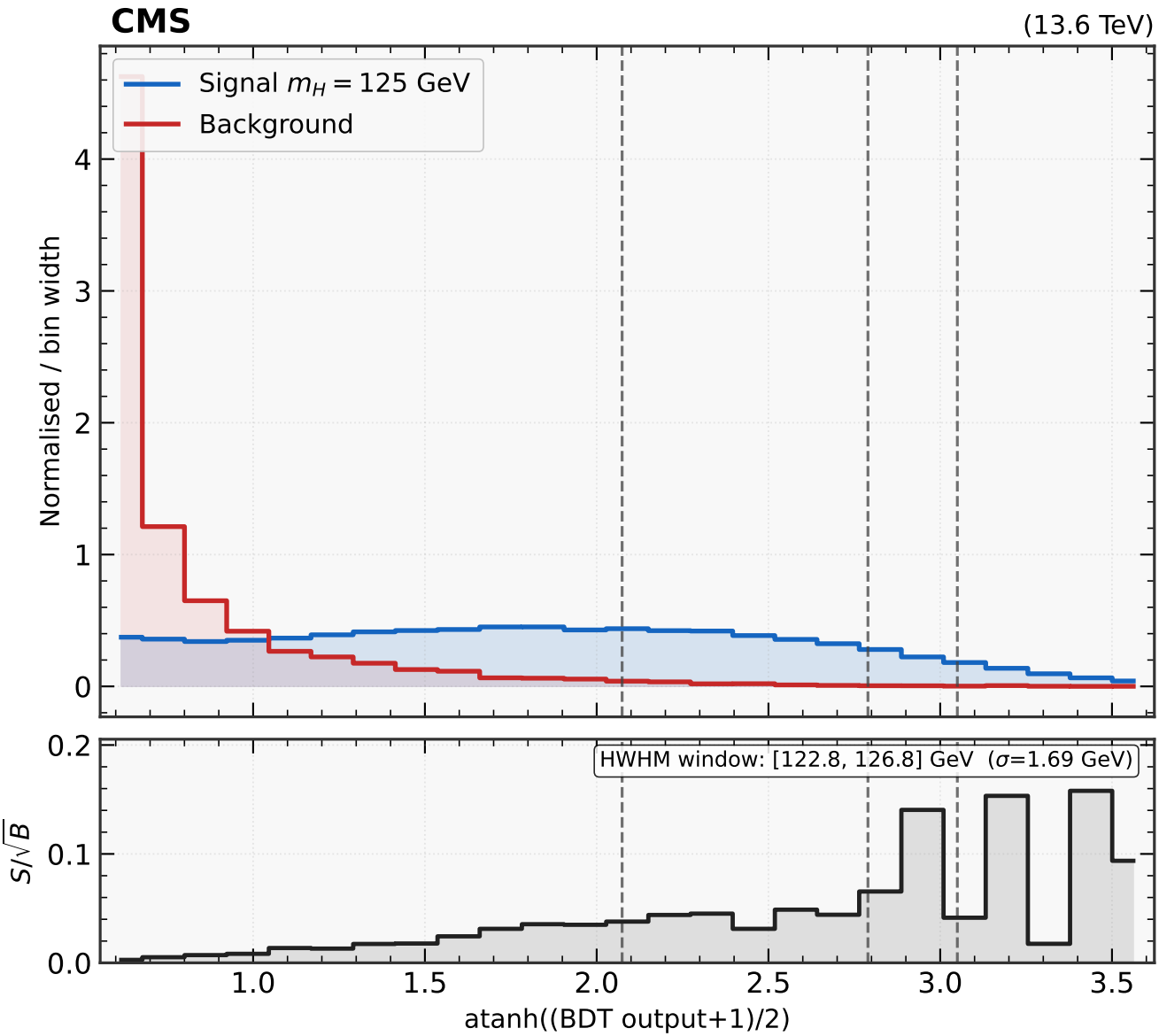

Combined significance vs. number of VBF BDT categories
CMS
(13.6 TeV)
Optimal: n=4, Z=0.2347
0.046
0.142
0.169
0.235
Combined Asimov Z [HWHM window]
0.050
0.075
0.100
0.125
0.150
0.175
0.200
0.225
1
2
3
4
Number of BDT categories

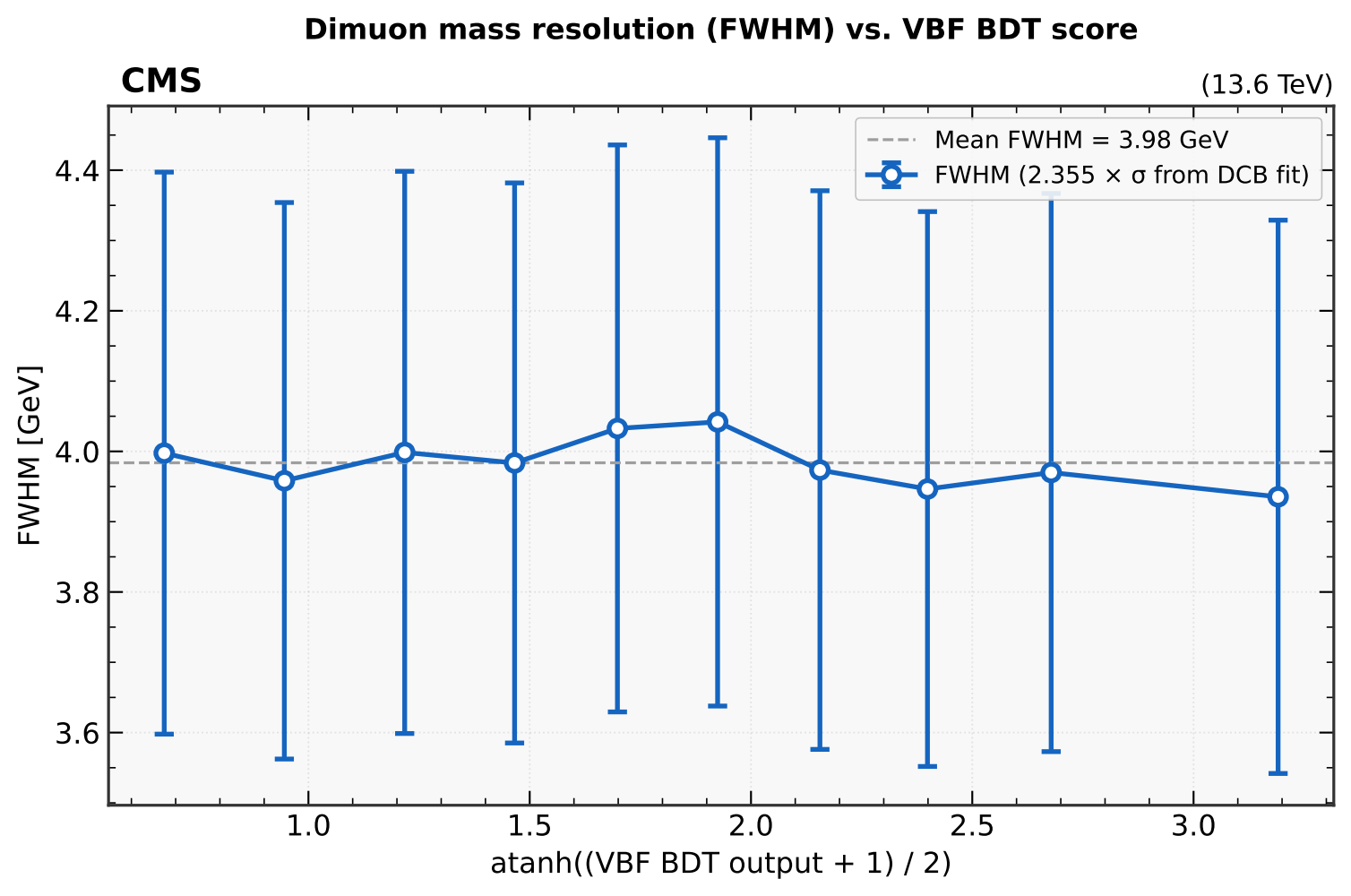
Iterative category boundary optimisation (AN Fig. 32 analog)

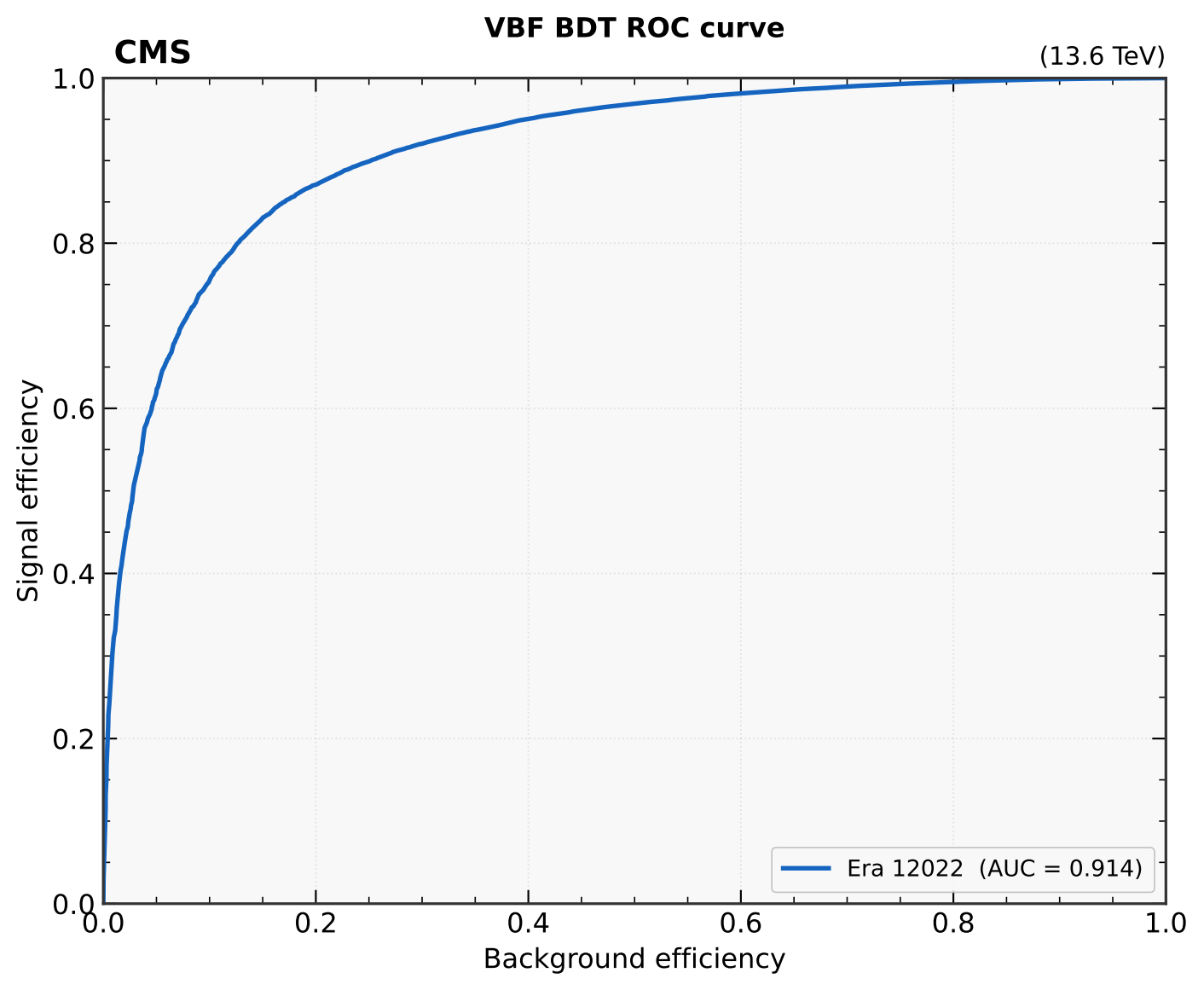


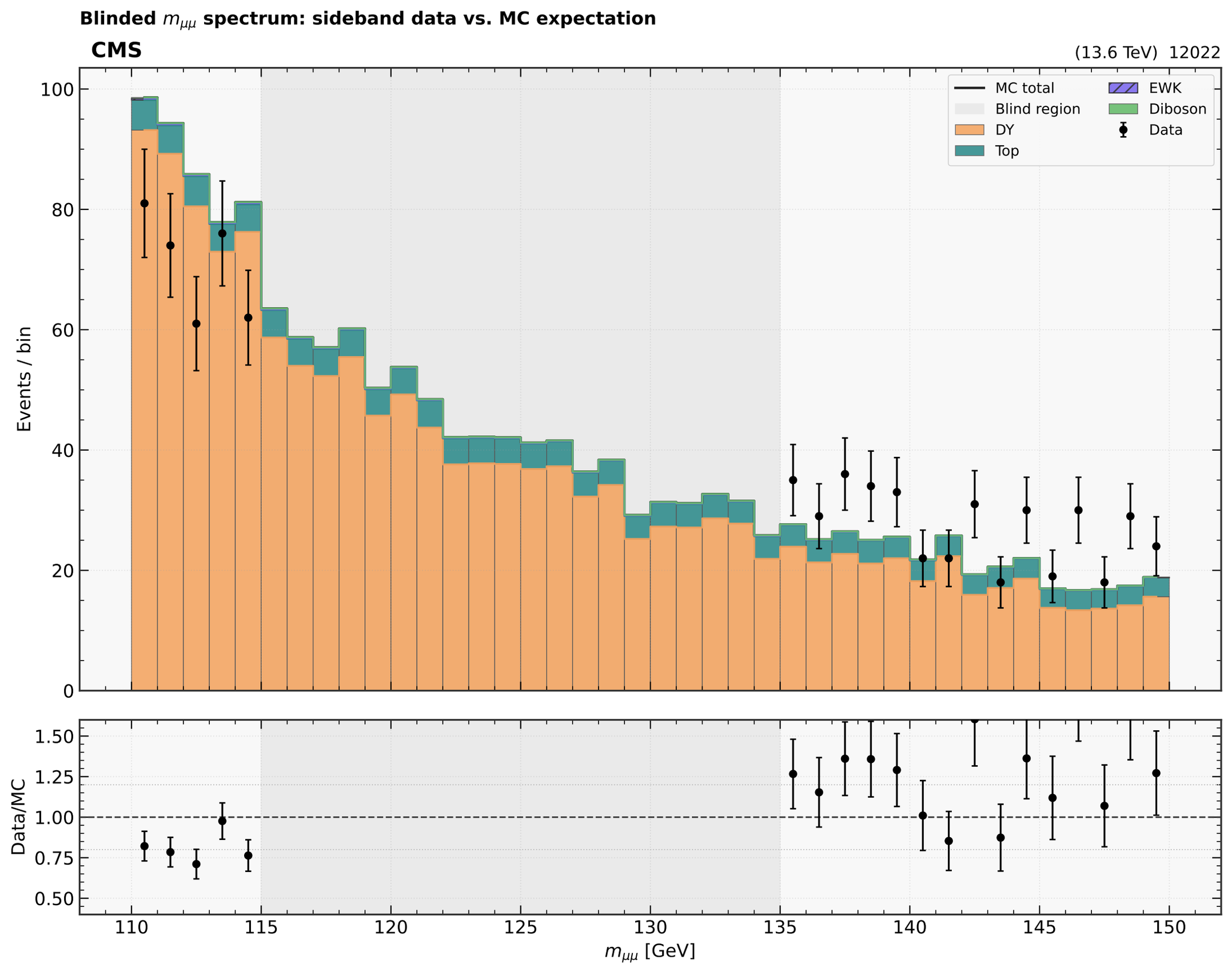


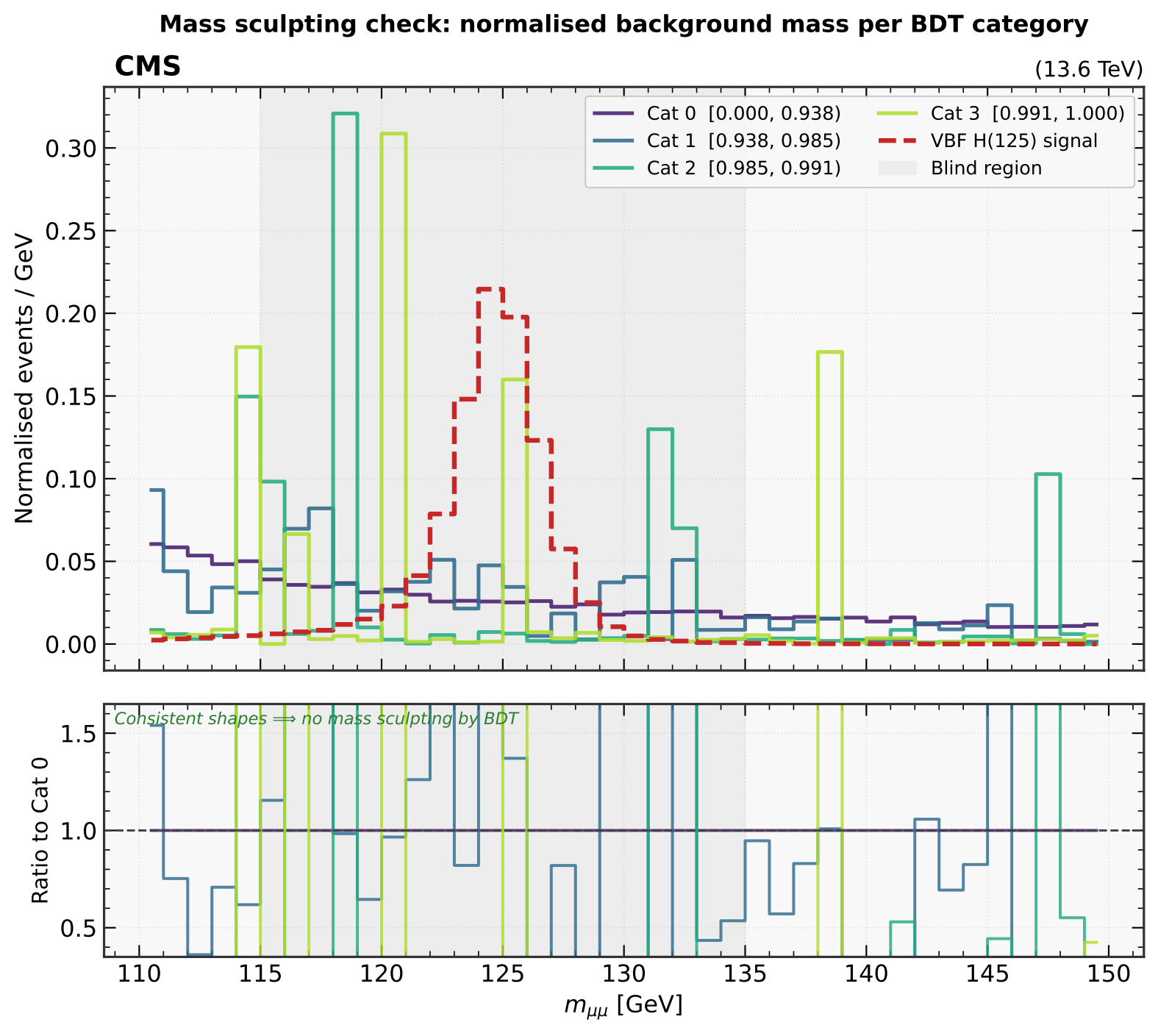
VBF BDT score and per-bin $S/\sqrt{B}$ in signal HWHM window




Dimuon mass resolution (FWHM) vs. VBF BDT score
CMS
(13.6 TeV)
Mean FWHM = 3.98 GeV
FWHM (2.355 × σ from DCB fit)
FWHM [GeV]
4.4
4.2
4.0
3.8
3.6
1.0
1.5
2.0
2.5
3.0
atanh((VBF BDT output + 1) / 2)

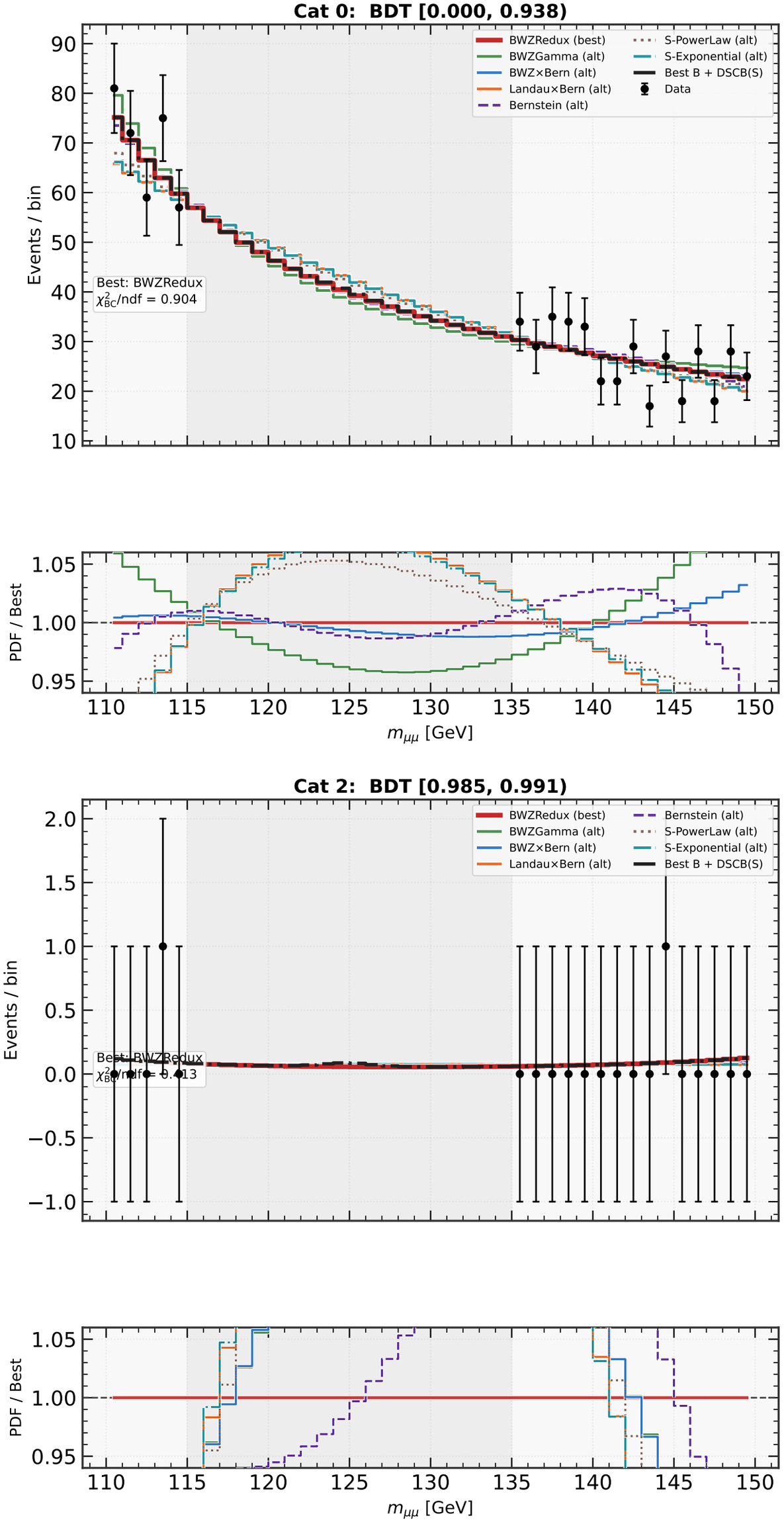

VBF BDT ROC curve
CMS
(13.6 TeV)
Signal efficiency
Background efficiency
0.0
0.2
0.4
0.6
0.8
1.0
Era 12022 (AUC = 0.914)


Blinded $m_{\mu\mu}$ spectrum: sideband data vs. MC expectation
CMS
(13.6 TeV) 12022
MC total
Blind region
DY
Top
EWK
Diboson
Data
Events / bin
Data/MC
$m_{\mu\mu}$ [GeV]

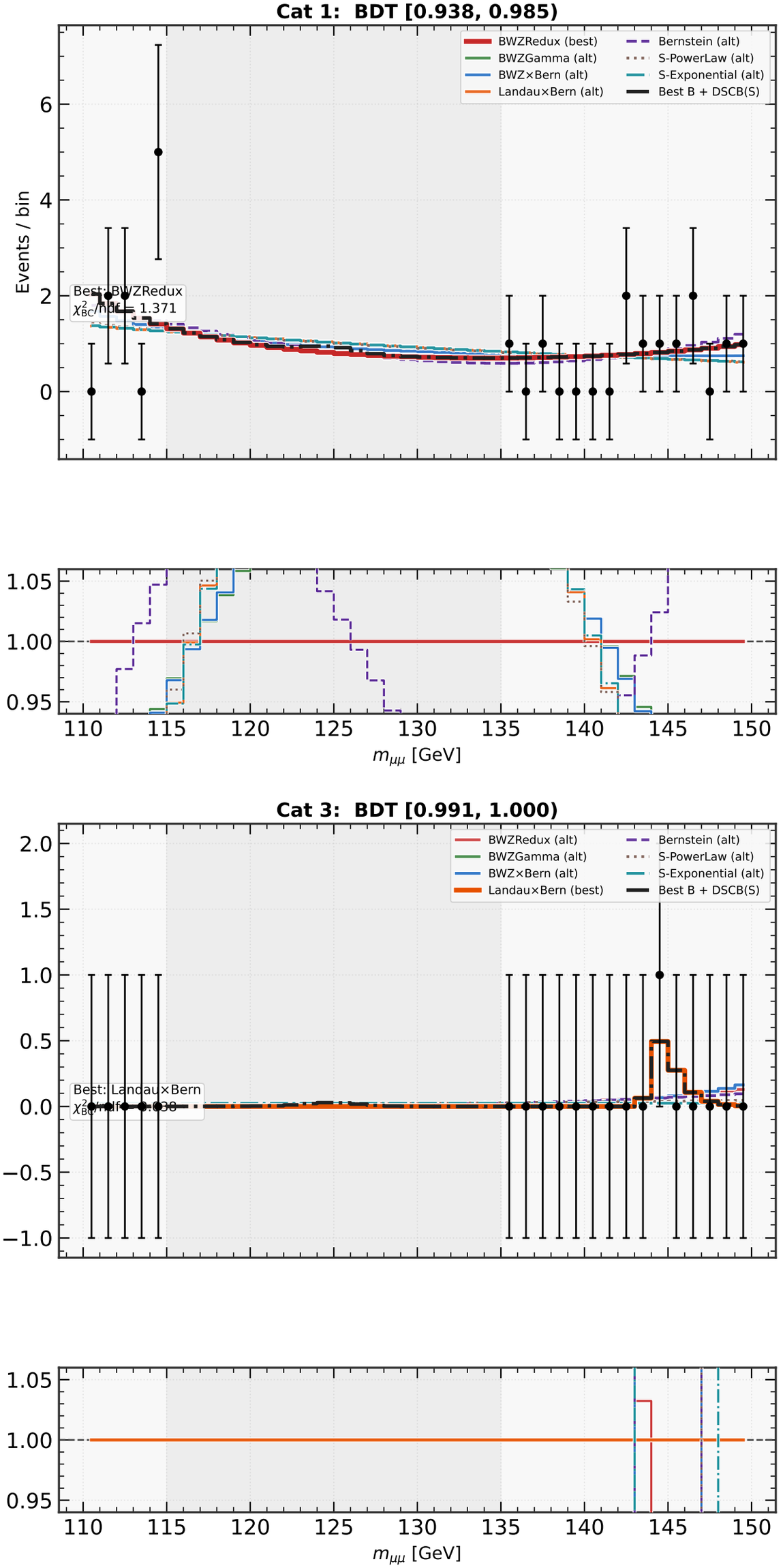

Mass sculpting check: normalised background mass per BDT category
CMS
(13.6 TeV)
Cat 0 [0.000, 0.938)
Cat 1 [0.938, 0.985)
Cat 2 [0.985, 0.991)
Cat 3 [0.991, 1.000)
VBF H(125) signal
Blind region
Normalised events / GeV
0.00
0.05
0.10
0.15
0.20
0.25
0.30
Consistent shapes ⟹ no mass sculpting by BDT
Ratio to Cat 0
0.5
1.0
1.5
110
115
120
125
130
135
140
145
150
$m_{\mu\mu}$ [GeV]

Per-category dimuon mass: background model fits

Cat 0: BDT [0.000, 0.938)



Cat 1: BDT [0.938, 0.985)



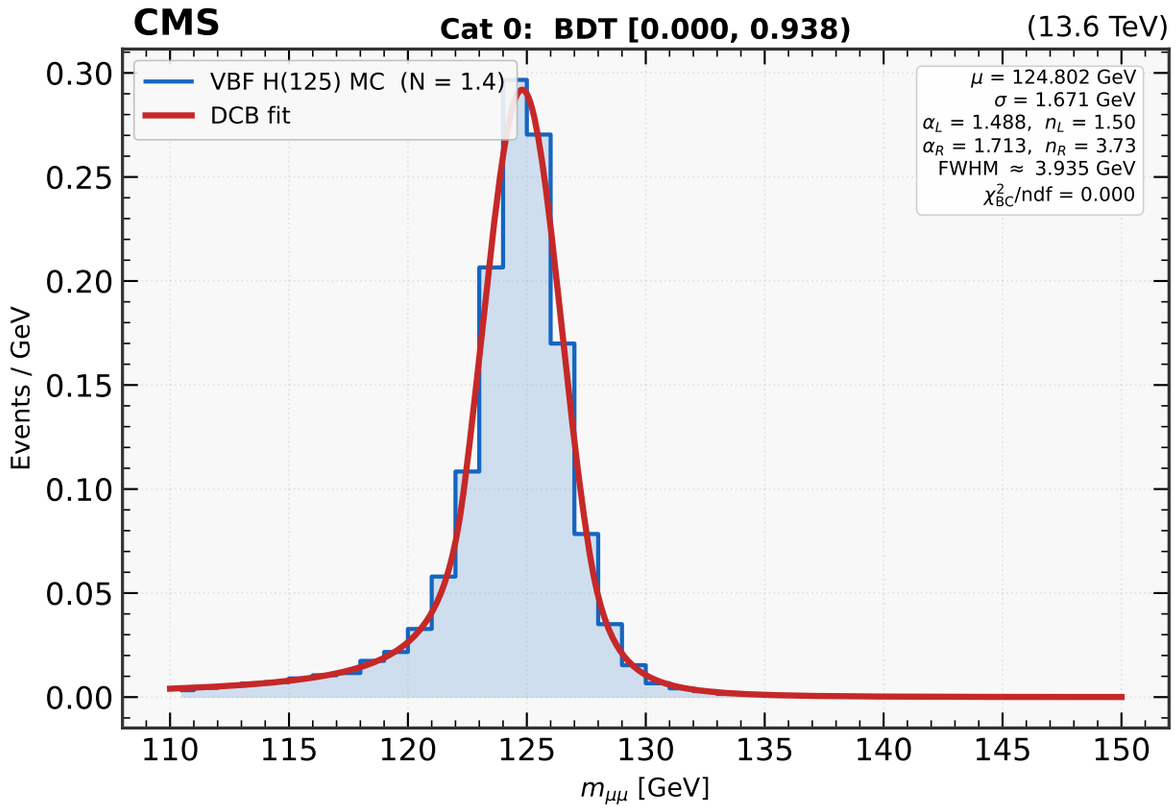
Cat 2: BDT [0.985, 0.991)

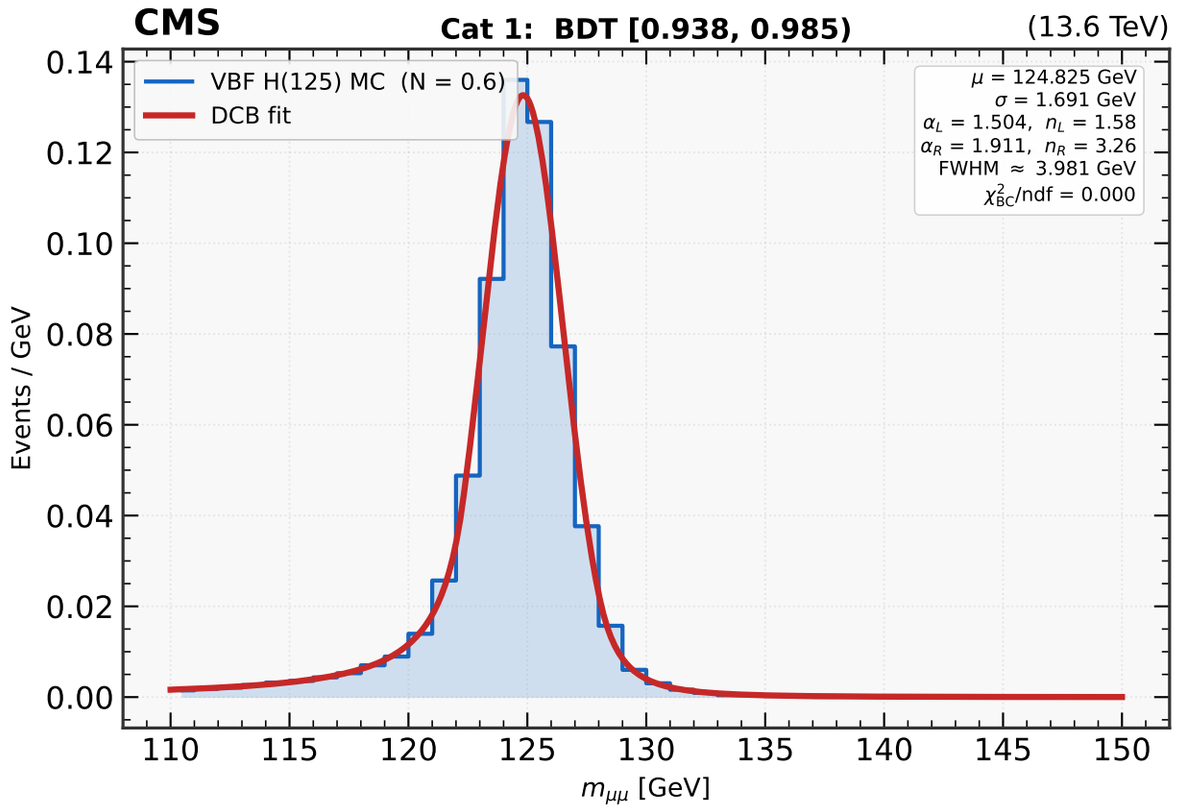
Cat 3: BDT [0.991, 1.000)

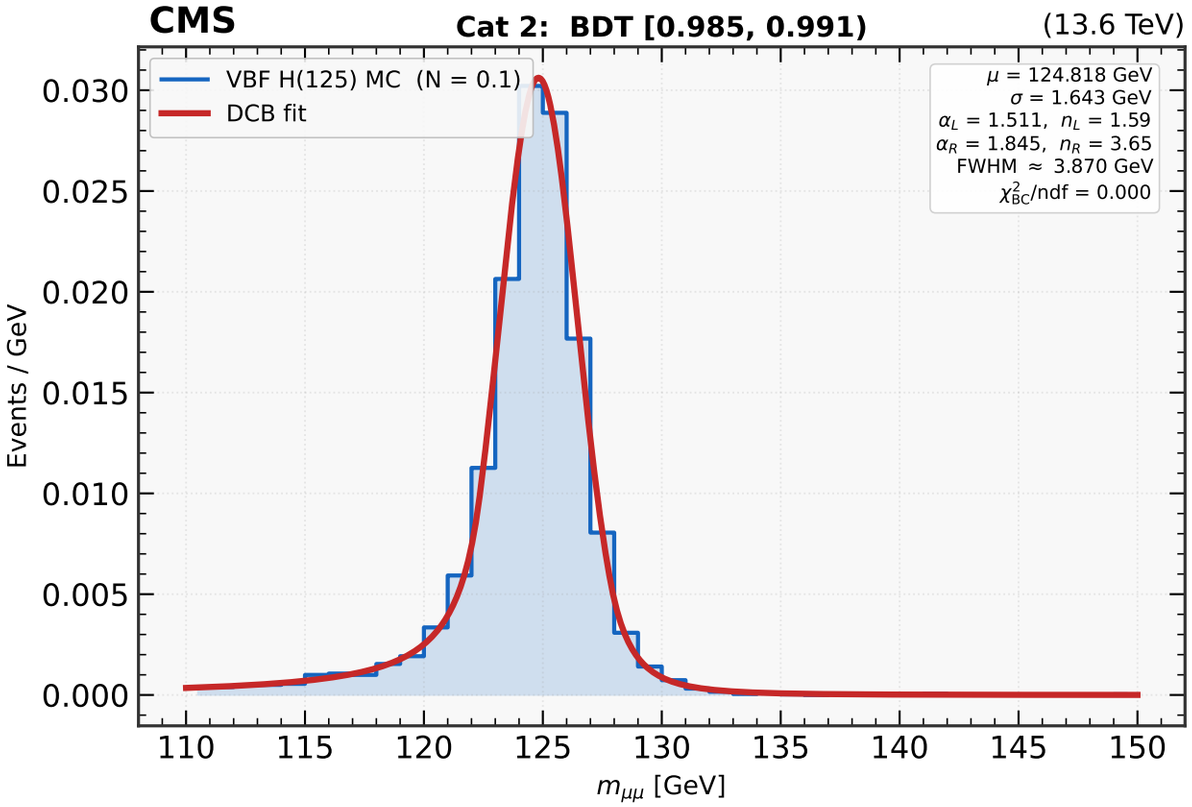
Per-category background model fit comparison (χ²_BC/ndf, ndf)

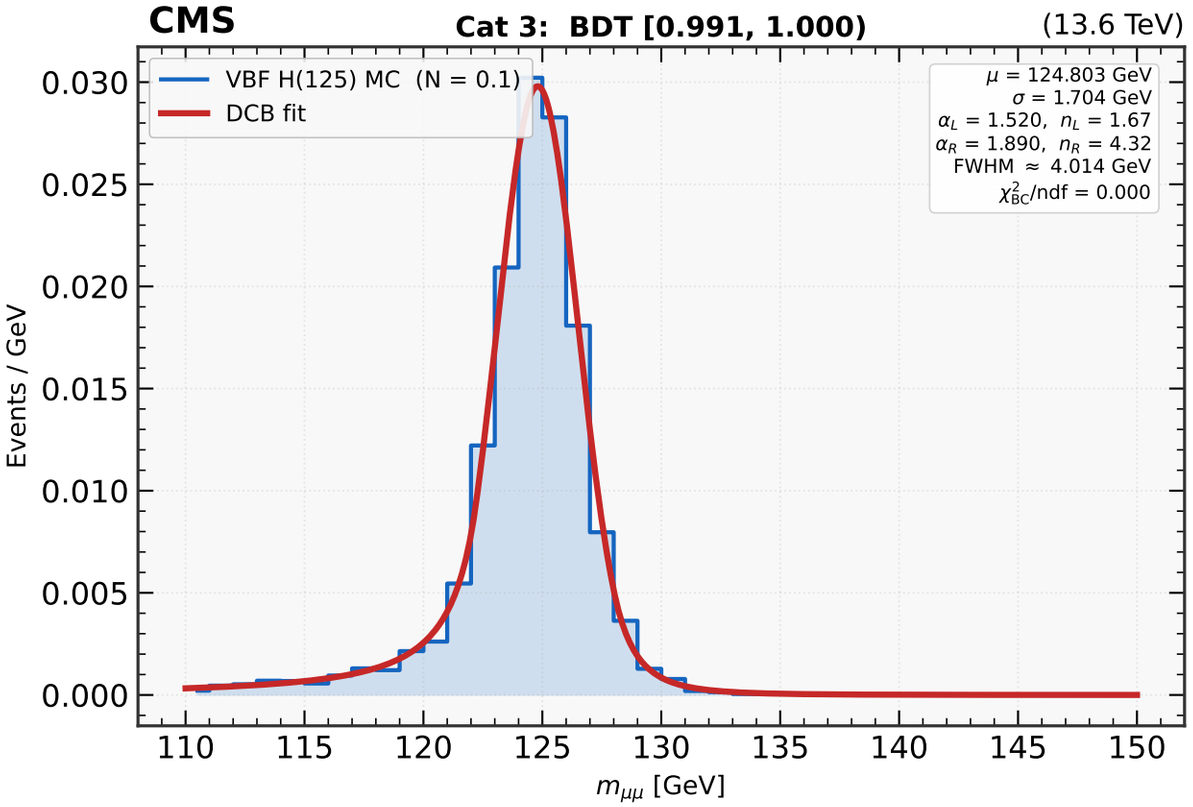
Category	BDT range	Best model	Best χ²_BC/ndf	BWZRedux	BWZGamma	BWZ×Bern	Landau×Bern	Bernstein	S-PowerLaw	S-Exponential
0	[0.000, 0.938)	BWZRedux	0.904	0.904 (37)	0.961 (37)	0.941 (36)	1.101 (37)	0.992 (35)	1.025 (37)	1.084 (37)
1	[0.938, 0.985)	BWZRedux	1.371	1.371 (37)	1.385 (37)	1.425 (36)	1.414 (37)	1.421 (35)	1.408 (37)	1.412 (37)
2	[0.985, 0.991)	BWZRedux	0.413	0.413 (37)	0.415 (37)	0.427 (36)	0.421 (37)	0.432 (35)	0.420 (37)	0.420 (37)
3	[0.991, 1.000)	Landau×Bern	0.038	0.140 (37)	0.146 (37)	0.150 (36)	0.038 (37)	0.157 (35)	0.175 (37)	0.199 (37)

VBF H(125) signal DCB extraction per BDT category



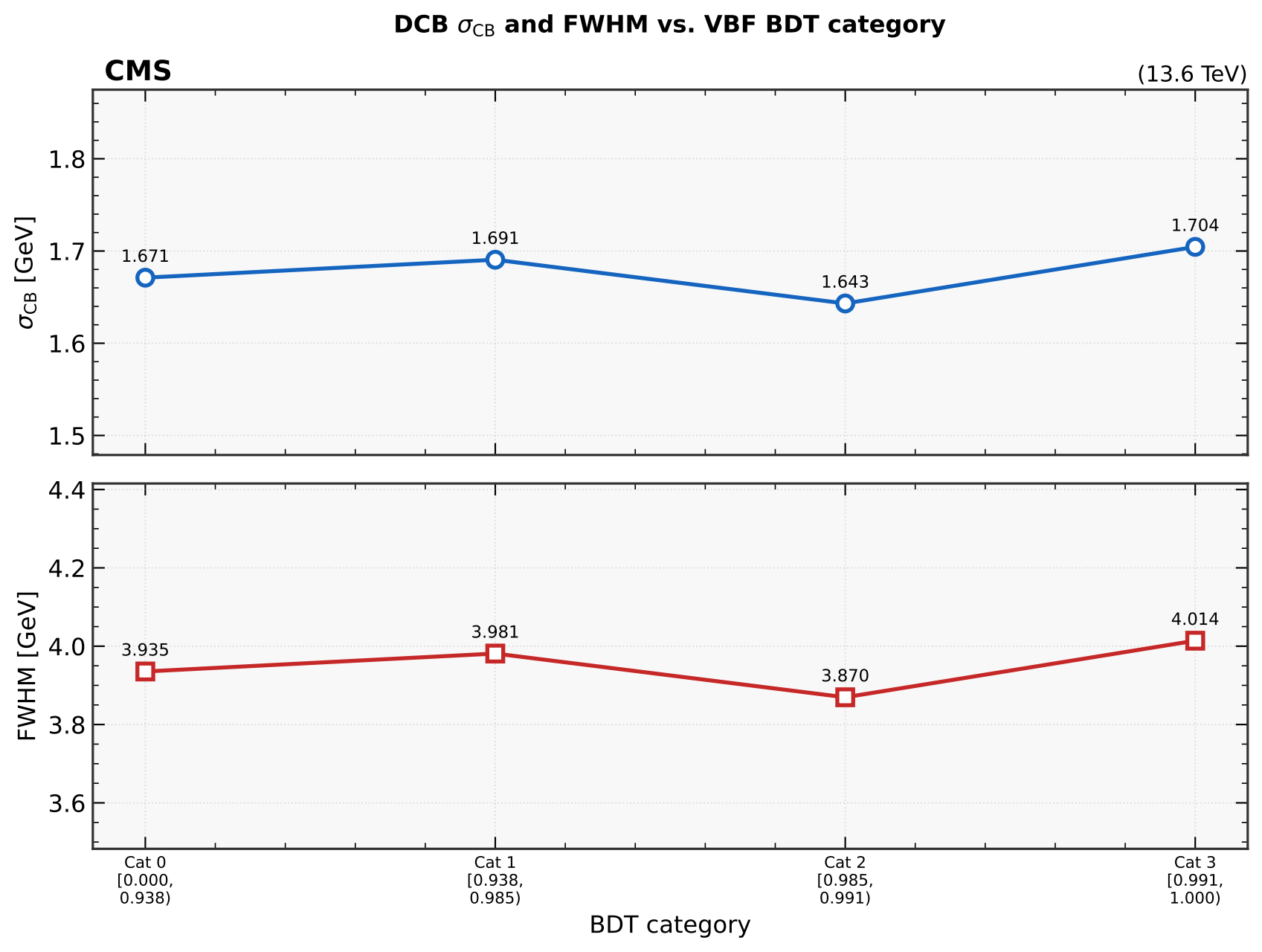


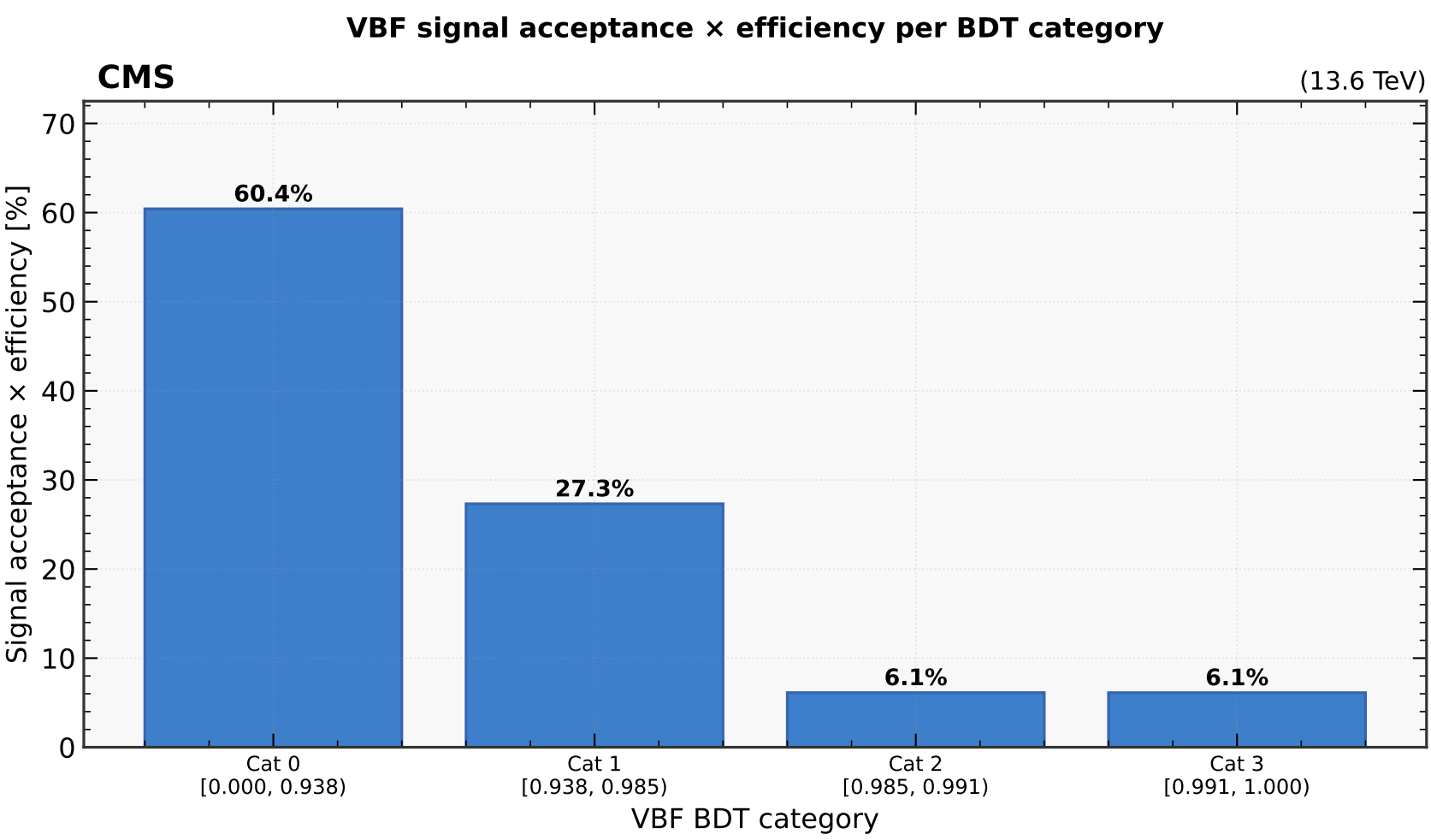




VBF signal DCB fit parameters per category

Category	BDT range	N_sig	Mean [GeV]	Sigma [GeV]	alpha_L	n_L	alpha_R	n_R	FWHM [GeV]	Asimov Z	χ^2_BC/ndf
0	[0.000, 0.938)	1.39	124.8024	1.6711	1.4882	1.496	1.7132	3.725	3.935	0.028	0.000
1	[0.938, 0.985)	0.63	124.8248	1.6905	1.5042	1.577	1.9114	3.255	3.981	0.102	0.000
2	[0.985, 0.991)	0.14	124.8179	1.6431	1.5111	1.589	1.8452	3.654	3.870	0.194	0.000
3	[0.991, 1.000)	0.14	124.8031	1.7045	1.5196	1.666	1.8897	4.321	4.014	0.071	0.000


DCB σ_{CB} and FWHM vs. VBF BDT category
CMS
(13.6 TeV)
σ_{CB} [GeV]
1.8
1.7
1.6
1.5
1.671
1.691
1.643
1.704
FWHM [GeV]
4.4
4.2
4.0
3.8
3.6
3.935
3.981
3.870
4.014
Cat 0 [0.000, 0.938)
Cat 1 [0.938, 0.985)
Cat 2 [0.985, 0.991)
Cat 3 [0.991, 1.000)
BDT category


VBF signal acceptance × efficiency per BDT category
CMS
(13.6 TeV)
Signal acceptance × efficiency [%]
0
10
20
30
40
50
60
70
60.4%
27.3%
6.1%
6.1%
Cat 0
[0.000, 0.938)
Cat 1
[0.938, 0.985)
Cat 2
[0.985, 0.991)
Cat 3
[0.991, 1.000)
VBF BDT category

Per-category yield and sensitivity summary

Category	BDT range	S_full	B_full	S_HWHM	B_HWHM	HWHM window [GeV]	S/√B	Asimov Z
0	[0.000, 0.938)	1.39	11307.86	0.938	1131.402	[122.8, 126.8]	0.028	0.028
1	[0.938, 0.985)	0.63	154.94	0.432	17.914	[122.8, 126.8]	0.102	0.102
2 ★	[0.985, 0.991)	0.14	13.33	0.095	0.212	[122.9, 126.8]	0.207	0.194
3	[0.991, 1.000)	0.14	11.37	0.098	1.901	[122.8, 126.8]	0.071	0.071

Category summary: HWHM-windowed signal and background yields

Category	BDT range	Data sidebands	MC sidebands ± stat	Data/MC (sidebands)	MC full (110-150) ± stat	S/√B [HWHM]	Asimov Z (cat) [HWHM]	B_HWHM
0	[0.000, 0.938)	741	5348.73 ± 82.77	0.139	11307.86 ± 120.28	0.0279	0.0279	1131.402
1	[0.938, 0.985)	20	53.90 ± 7.71	0.371	154.94 ± 13.92	0.1020	0.1016	17.914
2 ★	[0.985, 0.991)	2	4.20 ± 2.31	0.476	13.33 ± 4.12	0.2073	0.1941	0.212
3	[0.991, 1.000)	1	4.72 ± 2.80	0.212	11.37 ± 4.05	0.0714	0.0708	1.901
Combined	all						0.2320	

Category composition by process (full mass window)

Category	BDT range	DY	EWK	Top	Diboson	total_err2	total
0	[0.000, 0.938)	10158.58	20.83	1108.54	19.91	14466.81	11307.86
1	[0.938, 0.985)	136.69	7.09	10.76	0.40	193.76	154.94
2	[0.985, 0.991)	11.22	1.37	0.72	0.02	16.94	13.33
3	[0.991, 1.000)	10.01	1.11	0.23	0.01	16.41	11.37